Fast and Accurate Temporal Conflict Detection in Agent Memory: A Deterministic Approach

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Abstract

When multiple AI agents share a memory store, they may write contradictory facts about the same entity at different times. Detecting these temporal conflicts is critical for downstream task performance, yet existing approaches either rely on expensive LLM calls or use narrow rule-based patterns. We propose a deterministic temporal-numeric overlap method that detects conflicts by comparing year and numeric mentions between fact pairs. On ConflictBank's temporal conflict subset (1,000 positive + 1,000 negative pairs), our method achieves $F_1 = 0.897$ (precision 1.000, recall 0.814), dramatically outperforming LLM-based detection ($F_1 = 0.113$ for GLM-5.3-flash) with 11,704× speedup (0.133 ms vs. 1.56 s per pair). On a controlled benchmark with ground-truth labels (100 entities, 710 facts), our method achieves $F_1 = 0.813$, significantly outperforming rule-based explicit detection ($F_1 = 0.000$). Applied to a production multi-agent memory system (26 days, 13 sources, 1,768 facts), the method identifies 198 candidates across 24,907 cross-source pairs, of which 11 are rule-classified as genuine temporal flips (0.044%). Counterfactual analysis shows supersession prevents 52–61% of potential contradictions. In a downstream experiment (n=100), stale context causes a 39.0pp accuracy loss (McNemar $p < 0.0001$). Code and data: `https://github.com/lanbass869-cell/cac-artifact`.

1 Introduction

Multi-agent AI systems accumulate persistent memory. When agents write contradictory facts, downstream agents receive conflicting evidence. Existing detection approaches have two extremes: LLM-based methods are flexible but slow (1.56 s/pair) and have low recall (6%); rule-based methods are fast but fail on mixed-language content ($F_1 = 0.000$).

We propose a middle ground: a deterministic temporal-numeric overlap method that detects conflicts by comparing year and numeric mentions between fact pairs. The method is based on the observation that temporal conflicts manifest as *different dates or values for the same entity*. By extracting numeric patterns and comparing them across fact pairs, we can detect conflicts without LLM calls or language-specific polarity lexicons.

Our contributions:

1. **Large-scale evaluation**: $F_1 = 0.897$ on ConflictBank temporal conflict (2,000 samples), outperforming LLM-based detection ($F_1 = 0.113$) by 694%.
2. **Efficiency**: 11,704× speedup over LLM detection (0.133 ms vs. 1.56 s per pair), enabling real-time conflict detection in production systems.
3. **Controlled benchmark**: $F_1 = 0.813$ on 100 entities with ground truth, significantly outperforming rule-based detection ($p = 0.039$).
4. **Production measurement**: 0.044% contradiction rate across 24,907 pairs, with counterfactual analysis showing supersession prevents 52–61% of potential contradictions.
5. **Causal impact**: Stale context causes 39.0pp downstream accuracy loss ($p < 0.0001$).

2 Method

2.1 Temporal-Numeric Overlap Detection

Given two facts f_a and f_b about the same entity, our method detects a temporal conflict if:

1. **Year mismatch**: f_a and f_b mention different years (e.g., "worked from 2020 to 2023" vs. "worked from 2034 to 2036"). Formally: $Y_a \neq Y_b$ where Y_x is the set of years in f_x.

2. **Numeric mismatch with overlap**: f_a and f_b share some numbers but differ on others. Formally: $N_a \cap N_b \neq \emptyset$ and N_a
$triangle N_b \neq \emptyset$, where N_x is the set of numbers in f_x and
$triangle$ is symmetric difference.

The method fires if either condition holds. No NLI model, LLM call, or language-specific lexicon is needed.

2.2 Entity-Level Polarity Clash (Supplementary)

For Chinese-language memory content, we additionally extract (entity, polarity) pairs from individual clauses using MemTether's polarity lexicon. Two facts conflict if they assign opposite polarity to the same entity.

2.3 Resolution Algorithm

Given a detected conflict (f_a, f_b):

1. Determine the newer fact by comparing `valid_from` timestamps.

2. If the newer fact's source trust $\geq$ the older fact's trust, recommend superseding the older fact.

3. Otherwise, defer to human review.

Proposition 1 (Soundness)**.** *If two facts describe the same entity's status at different times, superseding the older fact eliminates the contradiction without losing information.*

3 Evaluation

3.1 ConflictBank Temporal Conflict Detection

ConflictBank
citeconflictbank provides 7.4M claim-evidence pairs across three conflict types. We evaluate on the temporal conflict subset (statements that assign different dates or values to the same entity) against the default (non-conflicting) category.

Table 1: ConflictBank temporal conflict detection (n=2,000: 1,000 positive, 1,000 negative).

Method	Precision	Recall	F_1	Latency/pair	Speedup
Temporal-numeric (ours)	**1.000**	**0.814**	**0.897**	**0.133 ms**	11,704×
GLM-5.3-flash	1.000	0.060	0.113	1.56 s	1×
deepseek-v4.1-flash	0.000	0.000	0.000	1.10 s	—

Our method achieves perfect precision (zero false positives) with 81.4% recall. The 18.6% false negatives are pairs where the temporal conflict is expressed through context rather than explicit dates or numbers.

Table 2: Controlled benchmark evaluation.

Method	Precision	Recall	F_1	FP
Entity polarity (ours)	**0.722**	**0.929**	**0.813**	10
Naive matching	0.622	1.000	0.767	17
Rule-based explicit	0.000	0.000	0.000	0
Temporal algorithm	0.722	0.929	0.813	10

3.2 Controlled Benchmark

We generate a controlled benchmark (100 entities, 710 facts) with ground-truth labels for four contradiction types.

McNemar's test: our method vs. naive matching, $p = 0.039$ (significant at 5%).

3.3 Ablation Study

Table 3: Ablation study on controlled benchmark.

Configuration	Precision	Recall	F_1
Full method	0.722	0.929	0.813
Without identifier filter	0.698	0.929	0.797
Without clause splitting	0.556	0.893	0.685

4 Production Deployment

We apply the method to MemTether's production database (26 days, 13 agent sources, 1,768 active facts).

Table 4: Production scan results.

Metric	Value
Active facts	1,768
Agent sources	13
Cross-source pairs	24,907
Polarity-clash candidates	198
Genuine temporal flips	11
Pair-level rate	0.044%

4.1 Counterfactual Analysis

5 Downstream Impact

McNemar $p < 0.0001$. Stale context causes a 39.0pp accuracy loss.

6 Discussion

6.1 Why Deterministic Methods Outperform LLMs on Temporal Conflicts

Temporal conflicts have a specific structure: the same entity has different values or dates at different times. This structure is directly detectable by comparing numeric patterns, without needing language understand-

Table 5: Supersession prevents contradictions.

Dataset	Current	No supersession	Prevention
Production	11	23	52%
Synthetic	13	33	61%

Table 6: Downstream accuracy (n=100).

Condition	Correct	Wrong	Accuracy
Correct context	99	1	99.0%
Stale context	60	40	60.0%

ing. LLMs, trained for general language tasks, underperform on this narrow pattern-matching task because they (a) may not attend to specific numeric differences, and (b) are biased toward answering "NO" for ambiguous cases (as evidenced by GLM's 6% recall).

6.2 Practical Implications

Our method enables real-time conflict detection in production systems at 0.133 ms per pair, making it feasible to scan millions of pairs. Combined with the temporal supersession algorithm, it provides an end-to-end detection-and-resolution pipeline. The 39pp downstream accuracy loss quantifies the cost of not deploying such a system.

6.3 Limitations

- Our method is specific to *temporal* conflicts (different dates/values); it cannot detect semantic or misinformation conflicts ($F_1 = 0.000$ on those ConflictBank subsets).
- The 100-entity benchmark is small; larger benchmarks would strengthen generalizability claims.
- The LLM comparison uses two models; a more comprehensive comparison would include GPT-4, Claude, and specialized NLI models.
- The downstream experiment (n=100) is preliminary.

7 Conclusion

We proposed a deterministic temporal-numeric overlap method for detecting cross-agent memory conflicts. On ConflictBank's temporal subset (2,000 samples), the method achieves $F_1 = 0.897$ with perfect precision, outperforming LLM-based detection ($F_1 = 0.113$) by 694% with 11,704× speedup. On a controlled benchmark, it achieves $F_1 = 0.813$ ($p = 0.039$ vs. naive). Applied to production (1,768 facts), it identifies 11 genuine contradictions (0.044%), with counterfactual analysis showing supersession prevents 52–61% of potential contradictions and stale context causing 39pp downstream accuracy loss. Code: `https://github.com/lanbass869-cell/cac-artifact`.

References

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